Stationary Vehicle Detection Using Complementary Information in Optical-SAR Images

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***Abstract*— Target detection in remote sensing images has received much research interest in the past years. Many saliency detection methods and constant false alarm rate (CFAR) detection methods have been proposed, which work effectively in simple scenes and high-resolution images, and unfortunately, they deteriorate in cases of complex scenes or low imaging quality. Multisource remote sensing images become much more accessible these years, and they provide redundant and complementary information about the area of interest. This article presents a detection method for stationary or slow-moving vehicle targets in complex scenes using optical and synthetic aperture radar (SAR) images. We utilize the contrast pyramid (CP) algorithm to fuse optical-SAR images and employ distinct methods to detect the saliency probabilities of optical, SAR, and fused images, respectively. Then, we calculate potential target areas based on these saliency probability maps and employ a region-growing method within these areas to extract the contours of possible targets. Finally, a CFAR detection method is used to obtain accurate vehicle targets. Experimental results demonstrate that the proposed method significantly improves the detection performance of stationary and slow-moving targets in complex scenes.**

***Index Terms*— Information fusion, saliency detection, synthetic aperture radar (SAR), target detection.**

I. Introduction

VARIOUS types of remote sensing images have become readily accessible with the advancement of geographic information technology. Target detection has been found very useful in many applications. However, obtaining clear remote sensing images consistently remains a challenging task due to complex scenes, subpar sensor performance, and limitations of imaging technologies. To prevent compromising detection performance due to degraded image quality, there is a necessity to derive a target detection method that can efficiently fuse imagery obtained from diverse sensors and facilitate complementary information integration.

Traditional target detection algorithms can mainly classify constant false alarm rate (CFAR) algorithms and saliency detection algorithms. CFAR algorithm [1], [2], [3] has been widely used in target detection of synthetic aperture radar (SAR), which utilizes the difference in pixel intensity between targets and background. Current research on CFAR primarily focuses on clutter statistical modeling and detector design. The dual-parameter CFAR method, which is based on the assumption of a Gaussian distribution, is one of the most classical clutter statistical models. However, it is more suitable for SAR images with low resolution and uniform clutter due to the assumption of Gaussian distribution [4], [5]. Quite a few improved CFAR methods based on non-Gaussian distributions, such as the Gamma distribution [6], Weibull distribution [1], G^0 distribution [7], K distribution [8], and generalized Gamma distribution [9], [10] have been proposed to deal with high-resolution and heterogeneous clutter scenarios. Gao et al. [11] combined the generalized gamma distribution and GAO models to provide a scheme for describing the statistical behavior of SAR pixel amplitude. Vitale et al. [12] designed an effective multiobjective cost function that accounts for texture, edge, and statistical properties preservation. Gao and Shi [13] found that the geometric perturbation polarization notch filter (GP-PNF) is suitable for sea clutter scenarios with a wide uniform range. The cell averaging CFAR (CA-CFAR) is a classical detector [14]. This method performs well in scenarios where there is only one target in the sliding window and a uniform clutter background exists. However, its performance rapidly deteriorates in the case of multiple targets. The suboptimal CFAR (SO-CFAR) was proposed in [15]. Nonetheless, SO-CFAR fails when all four detection windows contain target pixels. Rohling [5] therefore proposed the order statistic CFAR (OS-CFAR), which unfortunately exhibits poor performance in cases of uniform clutter. Gao et al. [7] introduced the adaptive cell-centered CFAR (ac-CFAR) detector, which can effectively handle the cases of uniform clutter and multiple targets. Further, Yu et al. [16] improved the traditional dual-parameter CFAR method by incorporating superpixels, resulting in the superpixel-CFAR method. This approach enables adaptive clutter region selection and demonstrates superior detection performance compared to ac-CFAR. Another CFAR technique, known as Global CFAR, was proposed in [17], which utilizes the entire image for obtaining detection thresholds during target detection, and its performance is sensitive to environmental changes. Liu et al. [18] proposed a novel simultaneous diagonalization transform (SDT) method that enables an almost simplest representation of information from the covariance matrix providing a rapid detection algorithm. Zhang et al. [19] reviewed the historical development and application status of compact polarimetric SAR, introduced its basic principle and the target detection approach, and summarized the currently proposed methods for compact polarimetric SAR target detection.

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Traditional CFAR algorithms suffer from a few aspects. First, it is difficult to select an appropriate clutter statistical distribution for target detection in complex scenes, while unmatched clutter models can lead to a significant decrease in detection performance. Second, a large number of artificial false alarms, such as buildings, can seriously affect the detection performance, and only calculating the intensity difference between the target and the clutter cannot eliminate these false alarms. Third, some targets have high contrast globally, but low contrast locally. If clutter regions cannot be effectively selected, it is difficult to detect such targets. Finally, due to the difference in pixel intensity in SAR and optical images, the CFAR detection method cannot be applied to optical images.

The saliency detection algorithm has been extensively used in optical images, particularly in non-remote sensing optical images. The early saliency detection model was proposed in [20] and [21], which can effectively detect spatial discontinuity in scenes. Liu et al. [22] first extracted a series of local, regional, and global features, including multiscale contrast, center surrounding histogram, and color space distribution, for the salient target, and then combined the features with conditional random field model to obtain the final saliency detection results. Jiang and Crookes [23], [24] reported a multilayer model based on successive Markov random fields (sMRF) and then discussed the visual significance of the image through its deep confidence propagation. This model can be optimized to obtain the final significance detection results. Cheng et al. [25] believe that reliable visual saliency estimation allows for appropriate processing of the target even without prior knowledge. Thus, a saliency detection method based on regional contrast was proposed, which considers both global contrast and spatial coherence, and achieves high precision and good recall. Perazzi et al. [26] obtained two saliency maps by calculating local and global uniqueness measures and distribution measures between regions, and finally assigned a pixel-level precision saliency value for each pixel based on high-dimensional Gaussian filter as the final saliency detection result. Jiang et al. [27] used a graph-based segmentation method to decompose the image into 15 different levels of segmentation results, then solved a regression problem to calculate a significance map for each segmentation result, and finally fused all significance maps to obtain the final detection result.

The traditional saliency detection method shows poor performance in remote sensing images [28], [29], especially SAR images, which is due to some factors: first, the scene of remote sensing images is much larger than that of nonremote sensing images, and the number of targets and interference in the scene is too large to distinguish. Second, the targets of interest in remote sensing images may appear anywhere in the image. As a result, the center prior and boundary prior are invalid for remote sensing images. Third, SAR images have speckle noise, which can seriously affect the detection results. To avoid these issues, some saliency detection methods for SAR images have been proposed, such as the classic data-driven detection method, pattern recursion (PR) [30], which performs intensity comparison at speckle level rather than pixel level. Some task-driven methods, on the other hand, combine scenarios to make judgments [31].

CFAR detection and saliency detection have their own advantages. However, they cannot effectively utilize multisource remote sensing data, which produces poor results under complex scenarios. Therefore, quite a few classic image fusion [32], [33], [34], [35] techniques have been developed, attempting to enhance detection performance in complex scenarios by improving image quality.

However, the primary emphasis of enhancing image quality via fusion algorithms remains on human visual perception. Since establishing a mathematical correlation between image fusion and target detection poses a challenge, effective evaluation parameters specific to detection performance remain elusive. Consequently, image fusion algorithms exhibit a degree of complementary information, which may not necessarily be advantageous for target detection. We will detail this issue in Section III and present comparative experimental results.

This article proposes an integrated algorithm for the fusion detection of stationary and slow vehicle targets in complex scenes, leveraging optical and SAR images. First, the saliency detection results obtained from optical, SAR, and their fused images are utilized as potential target regions. This approach, in contrast to the classical superpixel-CFAR method, retains only a small number of objects, therefore reducing the subsequent computational burden associated with clustering. Then, region growth is performed based on optical images with weak clutter to achieve clustering of possible targets. This strategy effectively preserves the contour of the targets. Finally, the guard bands are determined using the optical growth results, while the background bands are determined based on the region with the highest gray scale ratio on the optical image surrounding the corresponding guard band. This method enables accurate determination of the range of the background band and effectively prevents strong scattering targets, such as buildings, from falling within the background band.

This article is organized as follows. Section II presents the proposed fusion detection approach. Section III details the experimental results. Section IV concludes this work.

II. Fusion Detection Approach

The data used have achieved pixel-level accuracy registration using the V-DOG algorithm [36], and there is no significant change in the roadway of the scene. Due to differences in imaging mechanisms, there often exists a significant time difference for optical-SAR images acquired from the same platform, resulting in substantial displacement of high-speed moving targets in the optical-SAR images. Considering that fusion is meaningful only when the target positions in optical-SAR images are identical, the detection method proposed in this article and the utilized data exclusively pertain to stationary or slow-moving vehicles (vehicles with more than half of their positions overlapping in both images).

The process of the proposed fusion detection method is illustrated in Fig. 1 that includes the following steps.

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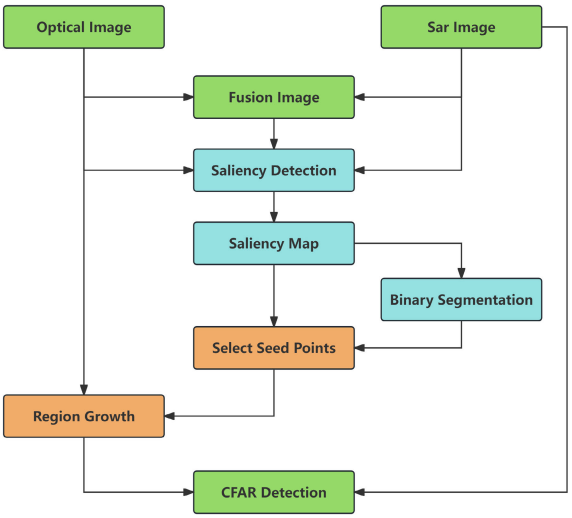


Fig. 1. Flowchart of the proposed fusion detection method for optical-SAR vehicle detection in complex scenes.

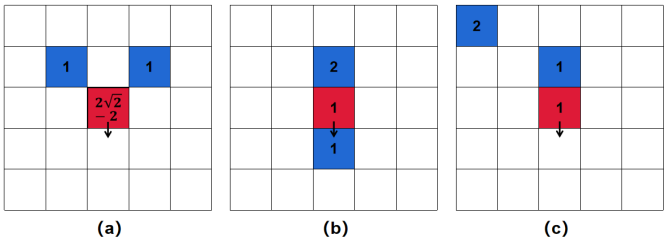


Fig. 2. Illustration of structural saliency superposition diagrams on three cases (a) distance between the PC points and the center are both $\sqrt{2}$, and the saliency radiation radius is 2. Therefore, the two points' radiations to the center point are $2-\sqrt{2}$, and the directions point to the lower left and right, respectively. The vector sum of the two radiations is $2\sqrt{2}-2$, and the direction is vertical downward. (b) Two PC points are opposite to the radiation direction of the center. The radiation intensity above is 2, and the radiation intensity below is 1. At this time, the vector sum of the radiations is 1, and the direction is vertical downward. (c) Distance between the PC point on the upper left and the center is $2\sqrt{2}$, which is larger than the saliency radiation radius, it is equivalent to (b) at this time.

1) *Generation of Saliency Maps:* Initially, we employed the contrast pyramid (CP) method for the fusion of optical and SAR images. Taking into account the differences among the optical, SAR, and fused images, we computed their corresponding significance probabilities utilizing three distinct saliency detection methods.
2) *Binary Segmentation:* We calculate the product of three saliency probability maps to describe the probability of the presence of targets in the scene and perform binary segmentation on them, retaining only regions with high saliency probability as possible target regions.
3) *Region Growth on Optical Image:* We select the pixel with the highest saliency probability in each possible target region as the seed point for that region and calculate the grayscale distribution within the region on the optical image. Finally, we choose the grayscale value with the largest distribution as the intensity of the seed point for that region and perform region growth from each point successively on the optical image.
4) *Generation of Background Bands:* We consider the multiple growth regions obtained in the previous step to be designated as guard bands. Subsequently, we generate rectangular quasi-background bands around each guard band, ensuring that their dimensions are twice the size of the bounding box enclosing the guard bands. We then proceed to calculate the grayscale histogram for each quasi-background band on the optical image. Pixels within the quasi-background band that exhibit grayscale values similar to the grayscale value with the largest distribution are retained, forming the final background band.
5) *Vehicle Target Detection:* All pixels within each protected region are detected based on the statistical model of the dual-parameter CFAR. If the proportion of detected pixels in the protected region exceeds a certain threshold T, which is defined as 30% in the proposed detection method, the region is considered a true vehicle target.

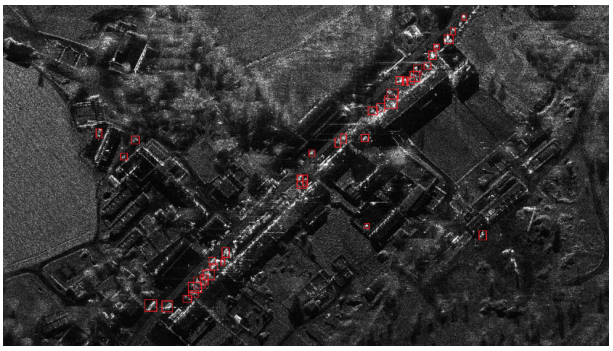
A. Generation of Saliency Maps

1) Optical-SAR Fusion: Previous research on fusion detection primarily focused on either a two-step approach of image fusion followed by detection or detection followed by decision fusion. Consequently, the development of fusion algorithms for either image or decision fusion becomes crucial. Unlike previous fusion-related papers, the fusion detection method proposed in this study aims to maximize the advantages offered by each image, such as the weaker clutter in optical images and the stronger target scattering in SAR images, and integrate them to enhance the detection results. The aim of the image fusion and association steps in this article is to acquire all potential target regions on the map. Consequently, the selection and innovation of image fusion algorithms do not significantly impact the detection performance of the proposed method. In this article, we use the classic CP image fusion method [37] to achieve this goal, while other classic fusion methods can also achieve the same result. The results of the several fusion algorithms and their impact on the method proposed in this article will be demonstrated in Section III.

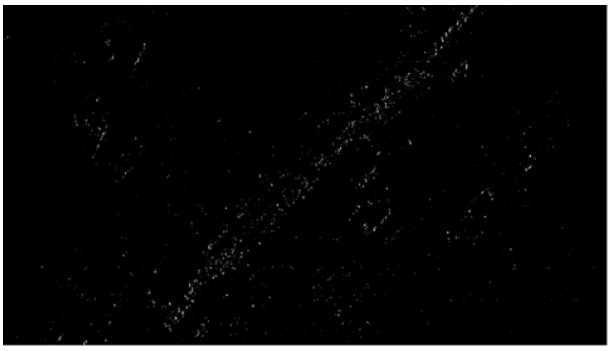
2) Saliency Detection: The objective of this step is to identify the locations of potential targets for subsequent clustering. In scenarios where targets are surrounded by other targets or buildings with similar intensities, it becomes challenging to distinguish them. Therefore, it is necessary to combine the detection results from multisource remote sensing images and their fusion image. Due to the significant differences in imaging principles between SAR and optical images, we computed the saliency probabilities using different saliency detection methods. For optical images, we employed an improved LTTI method for detection. This method selects the peripheral scale s and central scale c based on the prior knowledge of target size and image resolution. It calculates the difference of the feature maps corresponding to these two scales, thereby achieving center-peripheral difference operations. This approach offers advantages over the traditional LTTI method.

SAR image contains no color information, which degrades the performance of the LTTI method. We used the multi-blob's method [38] for saliency detection of SAR images.

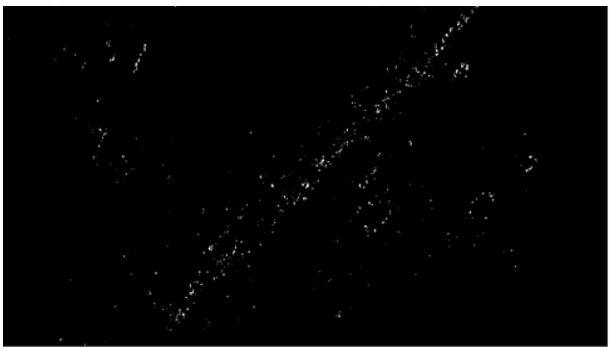
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(a)



(b)



(c)

Fig. 3. (a) SAR image. (b) Final saliency image. (c) Binary segmentation image.

Furthermore, considering that the fused image preserves spatial details, we applied the structural saliency superposition method for its detection. This method accurately identifies the center position of complex structures, which aids in selecting seed points for subsequent steps.

3) Structural Saliency Superposition Method: To mark the geometric center of the target, we propose a structural saliency superposition method. Specifically, the method is to generate a template block with the size of $(2R + 1) \times (2R + 1)$ with each point in the image as the center, and calculate the radiation of each point to the center in the area, respectively. We assume that there is a point A in the area and its phase congruency (PC) value is T, the radiation intensity of the point to the center O is $T \times K \times (R - L)$ (if the value is less than zero, it is equivalent to zero), where L is the Euclidean distance between A and O, K is the attenuation coefficient and set as 1 in this article, the radiation direction is the same as the direction of vector $\overrightarrow{AO}$. Finally, we calculate the vector sum of all radiations. For ease of understanding, we use three grid diagrams for display. Fig. 2(c) shows three blocks with the size of 5×5, and the attenuation coefficient is 1. The blue cells indicate the pixels in the block whose PC values were greater than the threshold, and the number indicates its PC value. The red cell represents the central pixel of the template block, the number represents the intensity of the vector sum of the radiation it received, and the arrow represents the direction of the vector sum. Particularly, although there is a point with a value in the upper left corner in C, the point has no radiation effect on the central pixel since its distance is greater than R. We use the same step to calculate the sum of the amplitude of the radiation at each point and calculate the ratio of the amplitude sum to the modulus of vector sum. Finally, we get the saliency radiation superposition map.

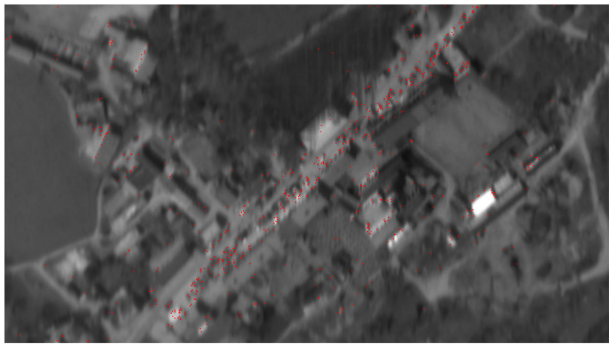


Fig. 4. Seed points on optical image.

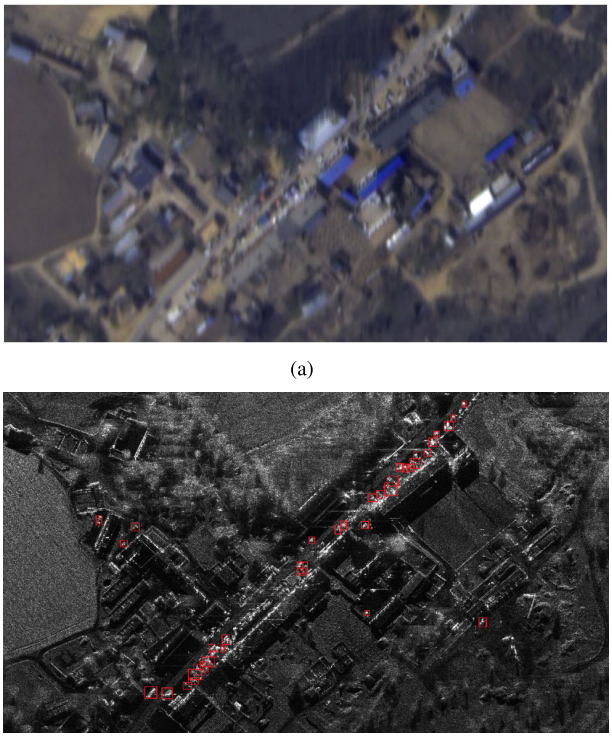
4) Threshold Segmentation: To obtain the final saliency map, we multiply the three saliency probability maps. The SAR image and the final saliency map are shown in Fig. 3(a) and (b), respectively. The vehicle target in the SAR image is marked with a red box. It is evident from the final saliency map that the significance probabilities of the target vehicles are consistently high.

We then calculate the intensity histogram distribution of the fused saliency map and obtain the cumulative distribution function T_{cdf}, and calculate the threshold for binary segmentation T_g from

$$1 - T_{\text{cdf}}(T_g) = \alpha \tag{1}$$

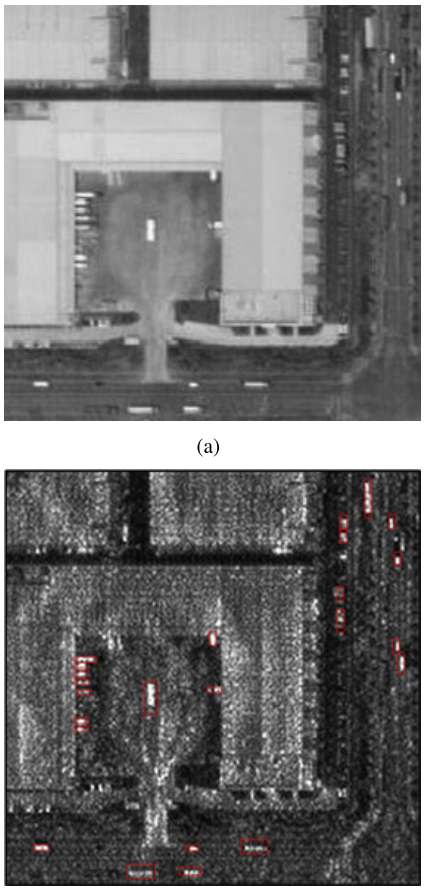
where $\alpha \epsilon [0, 1]$ is an empirical value that indicates the confidence level of a pixel being a target pixel. Due to the objective of this step being to acquire seed points for potential targets, only a few pixels are required for each target. This implies that we do not need to estimate the number and size of vehicles in the scene, as is necessary for traditional target detection algorithms. Even in scenes with numerous targets, such as parking lots, the value of the threshold will not exceed 5%. In this article, we set the value as 1%. It should be noted that increasing the confidence level does not affect the final detection results, but it does increase computational cost and time. Fig. 3(c) illustrates the binary image obtained after threshold segmentation.

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(a) (b)

Fig. 5. Real data. (a) Optical image. (b) SAR image.



(a) (b)

Fig. 6. Simulation data. (a) Optical image. (b) SAR image.

B. Region Growth Based on Optical Image

This step aims to obtain an outline of potential targets. In traditional region growth algorithms [39], a few pixels are chosen as seed points, and these regions gradually expand into larger areas. Specifically, this method starts with a set of initial seed points, selects neighboring pixel points that share similar properties with the seed points based on defined rules for region growth, clusters them together with their corresponding seed points, and continues this process until the termination conditions for region growth are met, forming the final region.

1) Selection of Seed Points: In the previous step, we segmented the final saliency map and obtained several disconnected regions. Considering that structural saliency detection has significantly increased the probability of saliency of the target center position, we selected the pixel within each region with the highest saliency probability as the seed point. In the cases where multiple pixels have equal saliency probabilities, we chose the pixel with the smallest Euclidean distance from the region's center as the seed point. As illustrated in Fig. 4, we marked the corresponding positions of the generated seed points on the optical image. Obviously, there is at least one seed point within each target. In particular, the position of the seed points is close to the center of the target. This approach ensures that the seed point's grayscale is not significantly different from the overall grayscale of the target, as pixels at the edge positions may correspond to the background's gray values.

2) Growth Rule: To determine the growth region, we set a threshold T_{gray} at 5% of the difference between the extreme gray values across the entire image, and use the grayscale difference between the seed point and its neighboring pixels as the criterion for judgment. Pixels with a grayscale difference less than T_{gray} are included in the growth region. When the grayscale differences between all the outermost pixels of the growing region and the seed point exceed T_{gray}, or when the outermost pixels are too far from the seed point (exceedingly twice the vehicle size), the region growth stops. In addition, we introduce a distance weight coefficient W_{distance}, which multiplies the grayscale difference calculation result, to ensure that growth can always succeed at very close distances while not altering the calculation results at longer distances. This effectively avoids growth failure caused by sudden changes in the grayscale of pixels surrounding the seed point during the initial growth stage. The weight coefficient W_{distance} is defined as

$$W_{\text{distance}} = 1./(1 + \exp(-(d - 3))) \tag{2}$$

where d is the Euclidean distance between the current pixel and the seed point.

C. CFAR Detection Based on Region Grow Results

The traditional CFAR detection methods encounter difficulties in accurately selecting background bands in complex scenes, making them susceptible to including nearby targets or other strongly scattered elements in the background band. As a

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TABLE I
AUCs OF DIFFERENT TARGET DETECTION METHODS AT THE SUPERPIXEL LEVEL

Image	CA-CFAR	Global-CFAR	Superpixel-CFAR	MSS	BMS	OSFD
ImageI	0.7111	0.5296	0.8398	0.9006	0.9277	0.9558
ImageII	0.3493	0.5019	0.6623	0.8555	0.7597	0.9128

result, the detection results are often unsatisfactory. To address this issue, we propose a method that ensures all pixels within the background band correspond to the actual background. In this article, we utilize several regions obtained from regional growth as guard bands and create a rectangular detection window around the geometric center of these guard bands. The size of the window is determined based on the guard bands. We calculate the grayscale values of all pixels within the detection window, excluding those in the guard bands, and identify the grayscale value with the highest proportion, along with its neighboring grayscale values, as the standard range. Subsequently, we select pixels with grayscale values falling within the standard range to form the background band. Furthermore, to mitigate interference caused by strong scattering points, such as building edges, we incorporate edge information from the optical image. Specifically, we eliminate all pixels corresponding to edges and their surrounding pixels within the background band.

After determining the guard band and background band, CFAR detection is applied to each pixel within each guard band. We adopt the classical dual-parameter CFAR method as the statistical model for clutter. It should be pointed out that for any pixel within the same guard band, the guard band and background band remain unchanged. Finally, we count the number of target pixels detected by CFAR in each guard band. If the number of target pixels exceeds 30% of the total pixels within the guard band, the region is classified as a target region.

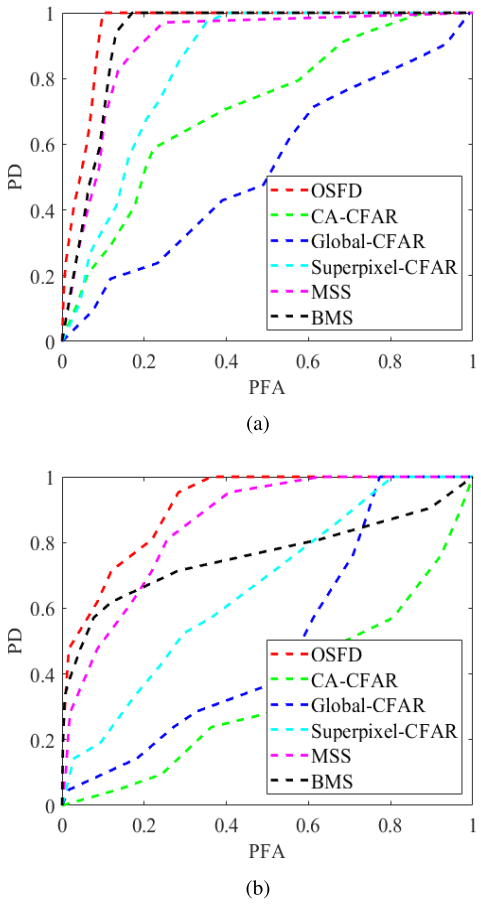


Fig. 7. ROC curves of different target detection methods for (a) real data and (b) simulation data at superpixel-level.

III. EXPERIMENTAL RESULTS AND DISCUSSION

A. Comparative Experiments With Traditional Methods

Due to the lack of SAR and optical images with a large number of vehicle targets in the same area, we selected a pair of airborne optical-SAR real data and a pair of simulation data to validate the method. Both sets of real data have sizes of 3701 $\times$ 2201 pixels. The SAR image of the simulation data was from by Chinese GF-3 sensor, and we used the cyclic Generative adversarial network [40], [41] to generate the corresponding pseudo-optical image based on the optical and SAR images captured at different periods. It should be emphasized that the real data is collected in rural township scenarios, where all vehicle targets are parked on both sides of the township roads or in courtyards without moving. Therefore, the vehicles in the optical images and SAR images are located in the same position. In the simulation data, the vehicle positions generated by the network remain unchanged, even if some of the vehicles are moving at high speeds. In Fig. 5, the real data are illustrated, while the simulation data are presented in Fig. 6. The vehicle targets in the SAR images are highlighted with red boxes. Note that the vehicles depicted in these images are the targets of interest, whereas the other prominent scattering points serve as clutter. The influence of surrounding buildings causes the vehicle targets to be submerged amidst strong clutter. Traditional target detection methods in SAR images aim to extract targets without a large number of false alarms.

The computation platform used is a workstation with a processor of AMD Ryzen7 7735H with a memory of 16 GB. We compare the fusion detection method respectively with CA-CFAR [14], superpixel-CFAR [17], global-CFAR [16], MSS [42], and BMS [43] detection methods. CA-CFAR, superpixel-CFAR, and global-CFAR are CFAR-based detectors, while MSS and BMS are saliency-based detectors. We quantitatively analyze and evaluate these methods at the superpixel level using two universal indicators of the receiver characteristic curve (ROC) [44] and the area under the ROC curve (AUC) [45].

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TABLE II
OPTIMAL OPERATION POINTS OF DIFFERENT TARGET DETECTION METHODS

Image		CA-CFAR	Global-CFAR	Superpixel-CFAR	MSS	BMS	OSFD
ImageI	PD	0.5882	0.619	0.7353	0.6765	0.8571	0.8571
	PFA	0.2231	0.554	0.2378	0.1028	0.1192	0.0832
	T_{loss}	0.6349	0.935	0.5025	0.4263	0.2621	0.2261
ImageII	PD	0.4762	0.2381	0.4286	0.8095	0.8095	0.9524
	PFA	0.6486	0.2711	0.2408	0.256	0.628	0.2831
	T_{loss}	1.1724	1.033	0.8122	0.4465	0.8185	0.3307

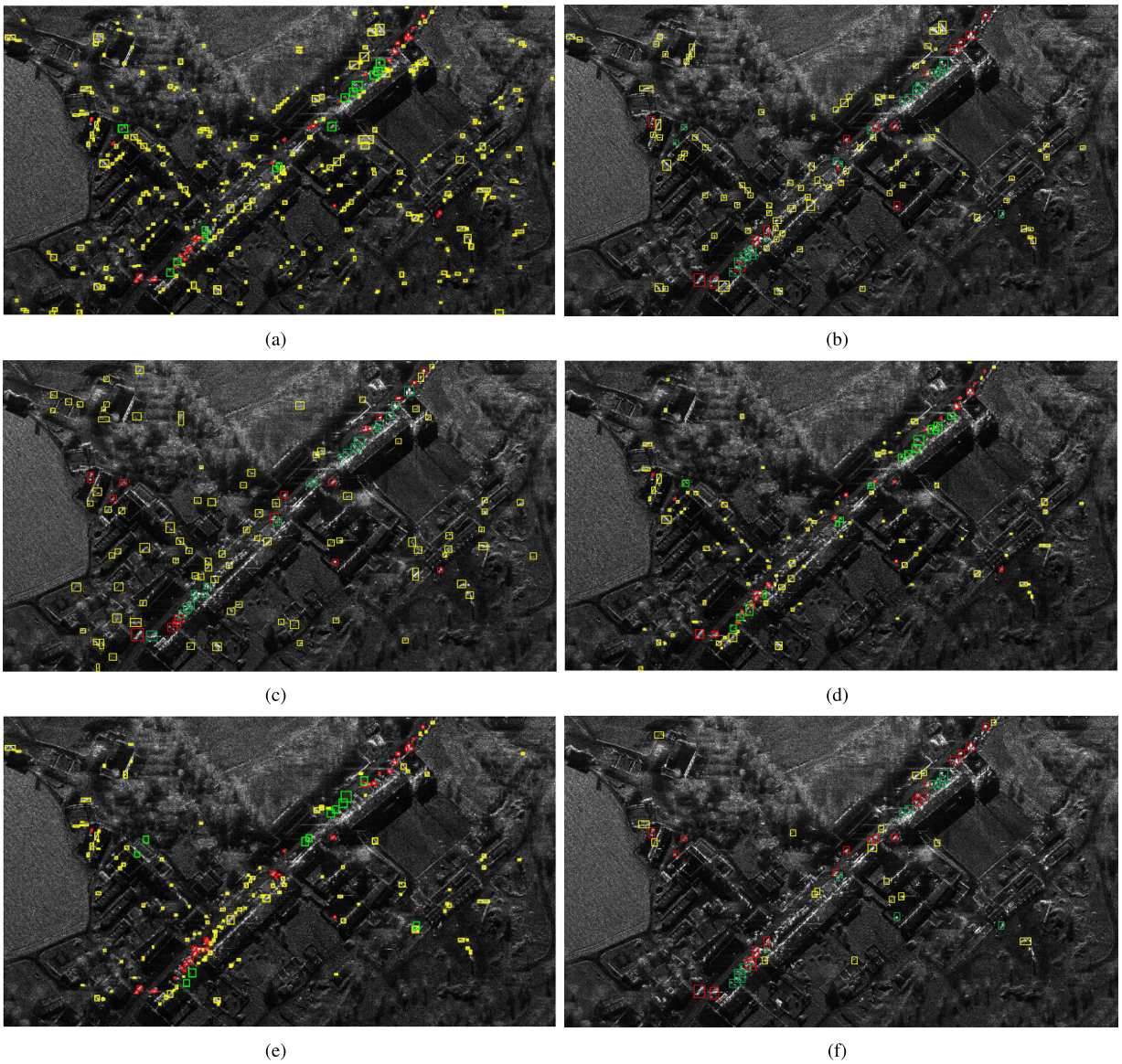


Fig. 8. Detection results of real data by (a) CA-CFAR, (b) Global-CFAR, (c) Superpixel-CFAR, (d) MSS, (e) BMS, and (f) OSFD.

Since vehicles are composed of multiple pixels in the image, superpixel-level statistics can better represent detection performance. Therefore, it is necessary to redefine PD and PFA based on the superpixel level. We define the set of superpixels where the target is located as $\{S_T(i)\}_{i=1}^{N_{TS}}$, where N_{TS} is the total number of superpixels of the target, and the set of superpixels corresponding to clutter is $\{S_C(i)\}_{i=1}^{N_{CS}}$, and N_{CS} is the total number of superpixels of clutter. The definitions of detection probability (PD) and false alarm probability (PFA) are given as follows:

$$\mathrm{PD} = \frac{N_{\mathrm{DT}}}{N_{\mathrm{TS}}} \tag{3}$$

$$\mathrm{PFA} = \frac{N_{\mathrm{dc}}}{N_{\mathrm{CS}}} \tag{4}$$

where N_{DT} is the number of target superpixels detected and N_{dc} is the number of clutter superpixels detected.

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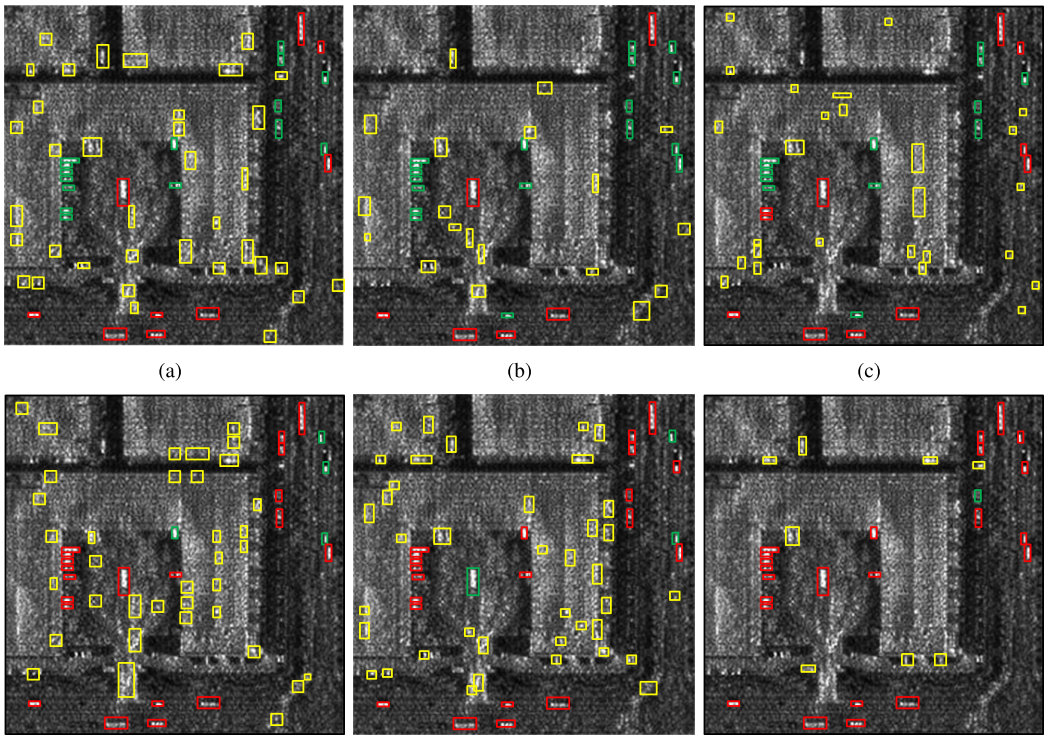


Fig. 9. Detection results of simulation data by (a) CA-CFAR, (b) Global-CFAR, (c) Superpixel-CFAR, (d) MSS, (e) BMS, and (f) OSFD.

Fig. 7(a) and (b) illustrate the ROC curves at the superpixel level respectively by different detection methods. For the sake of brevity and clarity in writing, we refer to the method proposed in this article as the optical-SAR fusion detection (OSFD) method. The proximity of the ROC curve to the upper-left corner indicates the superior performance of the corresponding method. It can be found that the proposed fusion detection method surpasses all other detection methods. The corresponding AUC values are illustrated in Table I. A higher AUC value indicates better performance of the detection method. The proposed detection method performs the best. Due to the complexity of the image scene, the information provided by SAR images alone is no longer sufficient for detection, and it becomes necessary to fuse information from heterogeneous sensors to improve detection performance.

We focus on the detection results at the optimal working point for discussion. The optimal working point is the point where the loss function reaches the smallest. The loss function is given by

$$T_{\text{loss}} = \text{PFA} + \beta(1 - \text{PD}) \tag{5}$$

where β represents the importance of false alarm losses and missed alarm losses. We set β to 1, indicating that their importance is the same. Table II shows the optimal operating points of different detection methods and their corresponding PFA and PD values.

Considering the known priors of image resolution and vehicle size range, we applied a morphological filter [43] to the detection results. This adjustment not only aligns the results with practical requirements but also facilitates their visual representation. Note that this filtering may lead to the removal of certain areas that contain targets. To ensure the fairness of the comparative experiment and avoid problems when calculating the PFA, all methods did not adopt filtering when drawing the ROC curve and calculating the AUC value of each method in the preceding paragraph. Figs. 8 and 9 illustrate the detection results of various methods at their respective optimal operating points for both real data and simulation data after the application of morphological filtering. Boxes with different colors are used, where the red box indicates the correct detection target, the yellow box represents a false alarm, and the green box denotes a missed alarm.

CA-CFAR and Global-CFAR perform the worst, exhibiting the lowest detection rate and highest false alarm rate. This outcome is entirely consistent with the results obtained from the ROC curve and AUC value analysis. The BMS method demonstrates favorable results when applied to real data, while its performance on simulation data is comparable to that of the Superpixel-CFAR method. The MSS method shows good performance on both datasets and surpasses the BMS method in terms of false alarm rate on the simulation data. The proposed fusion detection method proves to be the most effective, essentially because it successfully integrates information from SAR and optical images.

B. Comparative Experiments

Another advantage of the methodology proposed pertains to the seamless integration of fusion and detection processes. Previous methods typically perform image fusion first and then conduct target detection on the fused image, while our method

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(a)



(b)

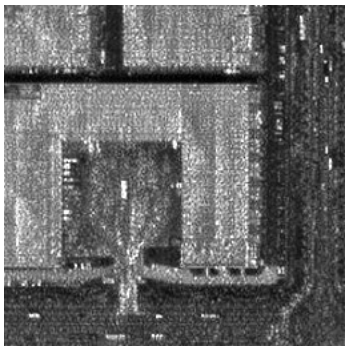


(c)

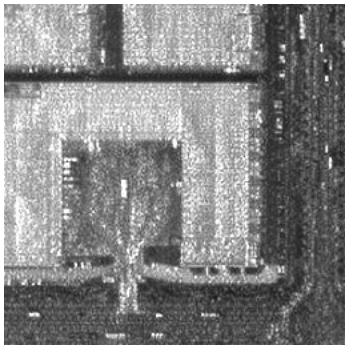


(d)

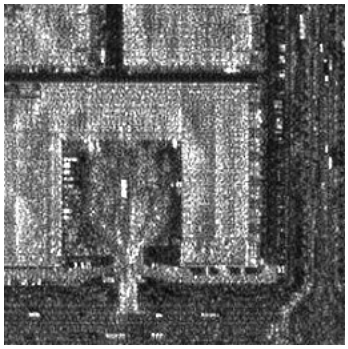
Fig. 10. Fused real images obtained by different methods. (a) LAP. (b) MST. (c) CP. (d) RP.



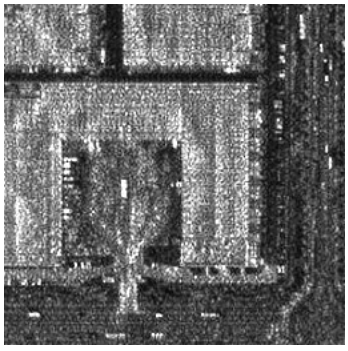
(a)



(b)



(c)



(d)

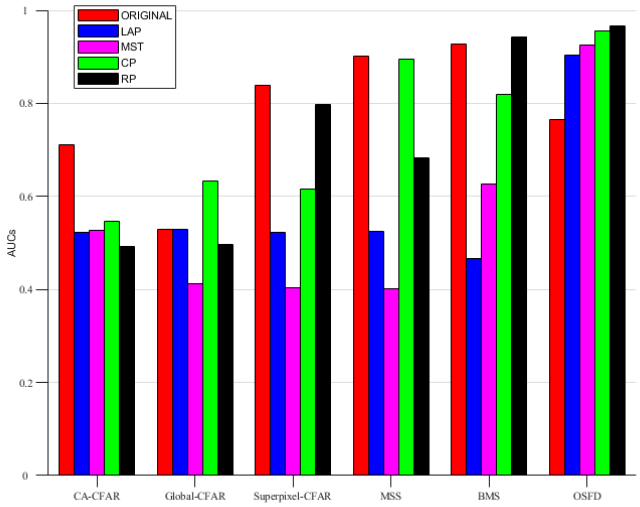
Fig. 11. Fused simulation images obtained by different methods. (a) LAP. (b) MST. (c) CP. (d) RP.

offers a more streamlined approach. In this study, we utilized four classic fusion algorithms [32], [33], [34], [35] to fuse the real and simulation data from the previous section and repeated the comparative experiments. The fused image is shown in Figs. 10 and 11. Subsequently, we used five detection algorithms to derive detection outcomes for both SAR images and the four fused images. Fig. 12 illustrates their AUC values. Specifically, the AUC value of original representation refers to the significance detection performed solely on optical and SAR images, with the results multiplied. The results of the OSFD

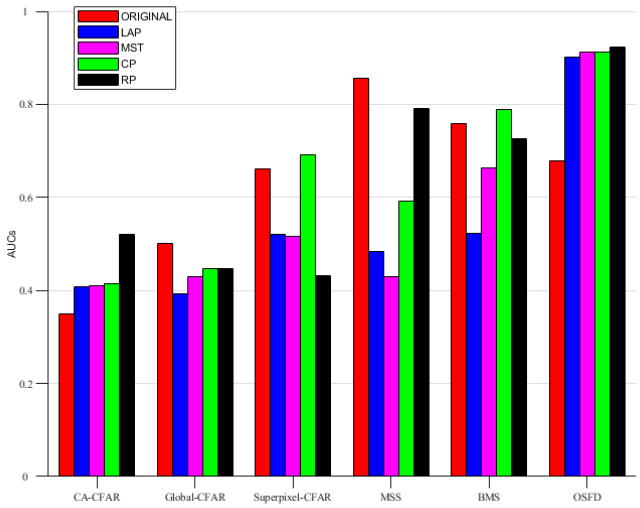
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TABLE III
IMPACT OF FUSION ALGORITHMS ON DIFFERENT DETECTION METHODS

Image		CA-CFAR	Global-CFAR	Superpixel-CFAR	MSS	BMS
ImageI	Minimum improvement	-0.31%	-22%	-51.9%	-55.3%	-49.6%
	Maximum improvement	-23.1%	16.3%	-5.1%	-0.6%	1.7%
	Compared to the proposed method	-42.8%	-33.8%	-16.6%	-6.4%	-1.3%
ImageII	Minimum improvement	16.6%	-21.8%	-35%	-49.7%	-31.3%
	Maximum improvement	49%	-10.8%	4.4%	-7.5%	3.9%
	Compared to the proposed method	-43%	-51%	-24.2%	-13.3%	-13.5%



(a)



(b)

Fig. 12. AUCs of various detection algorithms after improving image quality using fusion algorithms. (a) Real data. (b) Simulation data.

method were based on the fused image provided by the CP method.

As shown in Table III, it presents the maximum and minimum AUC value improvements achieved by various detection algorithms upon utilizing fused images. A negative enhancement denotes a diminution in detection performance. Additionally, a comparative analysis is provided to elucidate the efficacy of our proposed method.

It has been found that the image fusion algorithm may change the probability distribution of clutter in SAR images, which results in the collapse of model-based detectors, such as CFAR detectors. Consequently, despite the fact that fused images often adhere more closely to human visual preferences, detection performance may not undergo a commensurate enhancement. Although we can modify the parameters in the fusion algorithm to better align the fused outcome with detection methods, this endeavor generally amounts to adjusting the weights of optical and SAR images. Establishing a mathematically rigorous relationship between these weights and detection performance poses a challenge in interpretability. We can improve PD at the cost of increasing PFA when using CFAR, while the change in detection performance caused by increasing the weight of optical or SAR images may be non-monotonic. The proposed methodology does incorporate an image fusion step, it serves solely as a means of identifying potential target seed points, with subsequent detection relying primarily on the original images. As a result, the selection of the fusion method in our approach will not significantly compromise detection performance.

IV. CONCLUSION

This article presents a new approach for the detection of stationary and slow-moving vehicle targets in complex scenes based on optical-SAR data, aiming to improve traditional detection methods and also establishing the mathematical relationship between image fusion parameters and target detection performance. The experimental results of both the real and the simulation data have verified that the fusion detection method outperforms the classical algorithms.

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